

Scope of Work (SOW)

Development and Implementation of Integrated Renewable Energy Commercial, Billing, Settlement and Power Trading Management Platform

1. Introduction

SJVN Limited (SJVN), a Navratna Central Public Sector Enterprise, is currently acting as a Renewable Energy Implementing Agency (REIA) for the aggregation and supply of renewable power to DISCOMs and other procuring entities.

SJVN selects RE Developers (solar, wind, hybrid, Firm and Dispatchable Renewable Energy (FDRE), peak power, storage-linked projects, and Pumped Storage Projects etc.) for ISTS-connected projects across India through tariff-based competitive bidding. Subsequently, SJVN identifies suitable buying entities (primarily DISCOMs) for power offtake and executes back-to-back Power Sale Agreements (PSAs) with both developers and procurers.

SJVN currently manages an aggregated renewable energy capacity of approximately 6 GW, with plans to scale up to ~20 GW in the near to medium term. This scale-up is expected to significantly increase the volume, diversity, and complexity of commercial transactions.

To support this growth, SJVN intends to develop an Integrated Renewable Energy Commercial and Financial Management Platform. The platform will enable automated billing and invoicing across multiple contract structures, validation of energy data from grid operators, management of contractual provisions such as rebates, Late Payment Surcharges (LPS), and performance-linked adjustments, and structured handling of disputes, partial payments, and regulatory changes. It will also facilitate effective monitoring of receivables, payables, and payment security mechanisms, along with seamless integration with external and internal systems.

In addition to REIA-related commercial operations, SJVN also undertakes power trading activities including bilateral transactions and exchange-based trading across multiple power market products. SJVN intends to digitize and automate its power trading operations within the same RFP as an independent business module, covering exchange trading, bilateral transaction management, bid management, scheduling, NOC/PPA management, billing, settlement, reconciliation, regulatory reporting, client ledger, payment tracking, and management dashboards.

The proposed platform shall therefore comprise two independent business modules, namely the REIA Billing, Contract and Settlement Management System and the Power Trading Management System, along with a Management Dashboard and Consolidated MIS layer for senior management visibility across both business streams.

SJVN proposes to implement a centralized and integrated digital system to streamline and standardize its commercial operations in alignment with its expanding renewable energy portfolio. The proposed system shall be scalable, configurable, and adaptable to evolving regulatory and business requirements.



2. Objectives of the System

The objective of this assignment is to design, develop, implement, and operationalize a centralized, robust, and configurable digital platform to support SJVN's commercial operations related to renewable energy procurement and sale.

The proposed solution shall function as an integrated system to manage the end-to-end lifecycle of commercial transactions across SJVN's PPA and PSA portfolio, with a focus on enhancing process efficiency, accuracy, transparency, and control. The solution shall be vendor-agnostic, configurable, and aligned with industry best practices.

The key functionalities of the platform shall include:

- **Establish a unified commercial management system** to standardize and streamline processes related to contract management, billing, settlement tracking, and associated workflows, thereby creating a single source of truth across stakeholders
- **Ensure accurate and timely billing and settlement**, with consistent application of contractual provisions such as rebates, late payment surcharge (LPS), and other applicable adjustments
- **Strengthen financial visibility and control** through systematic tracking of receivables, payables, payment security mechanisms, and overall settlement status
- **Enable structured reconciliation and exception handling**, including identification, tracking, and resolution of discrepancies in a transparent and auditable manner
- **Improve governance, transparency, and auditability** by implementing role-based workflows, approval mechanisms, and comprehensive audit trails for all transactions
- Enable end-to-end digitization of power trading operations, including exchange trading, bilateral transaction management, bid creation, bid validation, bid submission workflow, scheduling, billing, settlement, reconciliation, and compliance reporting.
- Support trading operations across power exchanges including IEX, PXIL, HPX, and any other exchange or platform as required by SJVN.
- Support power exchange products including DAM, HPDAM, TAM, GDAM, RTM, GTAM, REC, ESCert, RPO-related products, and any other market products introduced by exchanges or regulators.
- Enable NOC, PPA, registration, client, generator, DISCOM, trader, and counterparty management for power trading operations.
- Provide automated and configurable validation for bids including pre-payment balance, margin requirement, NOC/PPA validity, quantum, time-block, and exchange-specific validations.
- Support billing, settlement, client ledger management, payment tracking, account reconciliation, digital signature, e-invoicing, and integration readiness with ERP/accounting systems.
- Provide market analytics, price forecasting, management dashboards, client dashboards, regulatory reporting, and consolidated MIS across REIA and power trading operations.



3. Scope of Work – Detailed Functional Scope

3A. Scope of Work - REIA Billing, Contract and Settlement Management System

The selected bidder shall be responsible for the design, development, implementation, testing, deployment, and post-implementation support of a centralized application for managing SJVN's commercial operations related to renewable energy procurement and sale.

The system shall support the end-to-end processes covering contract management, billing, reconciliation, settlement, dispute management and financial tracking across PPAs and PSAs.

The system shall support the following stakeholders:

- SJVN (Administrator and operational users)
- Sellers (RE Generators)
- Buyers (DISCOMs and other entities procuring power)

A. Stakeholder Onboarding and Registration

The system shall provide functionality for onboarding of all participating entities including sellers and buyers.

The onboarding module shall support the registration and management of the following entities:

- Sellers (RE Generators)
- Buyers (DISCOMs and other entities procuring power)

This Onboarding should include all necessary information related to the stakeholders but not limited to the following:

(i) For RE Generators:

Generation Capacity, Plant details, Organizational Details and required regulatory approvals & other supporting documents etc.

(ii) Buyers (Discoms/C&I/Other Power Purchasing entities)

Contracted/Installed Capacity

PSA Tariff

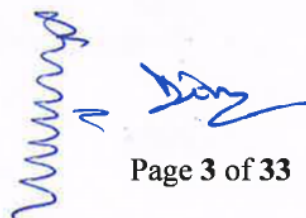
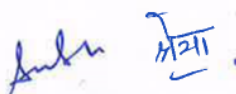
Criteria for Supply of Power

Regulatory approvals and other supporting documents etc. if any.

B. Contract Management

The system shall provide functionality for managing energy procurement and supply contracts between:

- SJVN and Sellers



- SJVN and Buyers

Key capabilities include:

- Creation and Management of short-term and long-term energy contracts
- Capture of contract details including capacity, tenure, tariff, billing rules, payment terms, and security mechanisms.
- Monitoring and Management of validity, expiry, renewal of EMD & PBG (BG/ISB/POI etc.)
- Support for multiple contract types such as solar, wind, hybrid, FDRE, peak power, PSP, storage-linked arrangements and other Types of Projects as per the future market trends.
- Provision for contract amendments, version control, and maintenance of revision history.
- Version management and audit logs
- Bulk upload of contract data through predefined templates including EMD/PBG data.
- Contract search, filtering, and reporting features

C. Stakeholder Profile Management

Both sellers and buyers shall have the ability to manage and update their profile details subject to approval by the SJVN.

Capabilities shall include:

- Registration and approval workflows for developers/sellers and buying entities (DISCOMs, etc.)
- Capture of organizational details, regulatory approvals, banking information, and contact details
- Document management for agreements and supporting records
- Profile update functionality with audit trail

All profile updates shall be recorded with complete audit history.

D. Energy Data Accounting & Validation

The system shall support capture and validation of energy data required for billing and settlement. The system shall support handling of both provisional and final energy data, including validation, revision tracking, and locking of data post-finalization.

Scope includes:

- Integration with external data sources (e.g., SEA/REA/RLDC/SLDC/JMR), where available and required
- upload of provisional energy data & final energy data
- Provision for manual upload, validation of energy data through Auto fetch from REA data.
- Storage of periodic energy data (time-block or aggregated) and block-wise energy data

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- Validation of energy data against contractual parameters and identification of exceptions
- project-wise energy allocation
- Detection of energy deviations
- Energy validation reports

Once energy data is finalized it shall freeze.

E. Billing and Invoicing

The system shall enable SJVN to provide controlled access to Sellers for creation and submission of invoices through the platform. Upon submission, the invoices shall be processed through a defined workflow for review and approval by SJVN.

Post approval, the system shall support application of applicable deductions, charges, and adjustments as per contractual provisions, including payment security-related adjustments and other applicable components.

Subsequently, the system shall facilitate generation of corresponding invoices to the Buyers, incorporating approved energy quantities, applicable charges, adjustments, and SJVN's trading margin, in accordance with the relevant contractual terms.

The overall billing process shall be configurable to accommodate different contract structures, billing cycles, and evolving business requirements.

Scope includes:

- Automated invoice generation based on PSA and PPA obligations, Contract tariff, Actual energy supplied, Applicable charges, Energy charges, Transmission charges (if applicable)
- Support for two-sided billing (developer and buyer)
- Configurable billing cycles (Monthly, weekly, daily or custom)
- Generation of provisional invoices based on provisional energy data
- Provision of managing the Disputed and Un-Disputed Billing amount
- Generation of final invoices based on finalized energy data
- Generation of supplementary invoices for adjustments arising due to tariff revisions, change in law, late payment surcharge, energy revisions, or contractual changes
- Inclusion of taxes, statutory charges, and other applicable components in invoice computation
- Invoice structure to include unique invoice number, due date, and detailed charge breakup
- Invoice generation through defined approval workflows
- Invoice versioning, storage, and retrieval
- Automated sharing of invoices and notifications

F. Seller Invoice Management

The system shall provide functionality for submission and management of invoices from developers.

Scope includes:

- Upload of seller invoices using predefined templates
- Validation of uploaded invoice data against system-generated values
- Storage and tracking of seller invoice submissions
- API-based integration for automated invoice submission (where applicable)
- Status tracking of submitted invoices

G. Invoice Approval, Distribution and Notification Workflow

The system shall provide a configurable workflow for invoice validation, approval, and automated distribution to the respective buyers.

Before dispatch, all generated invoices shall pass through a predefined approval workflow to ensure accuracy, compliance with contractual terms, and proper authorization by designated officials.

The workflow shall support the following capabilities:

Approval Workflow

- Multi-level approval hierarchy configurable as per organizational roles.
- Role-based approvals for invoice validation and authorization.
- Workflow status tracking for each invoice throughout the approval lifecycle.
- Complete audit logs capturing user actions, timestamps, and approval decisions.
- Provision for rejection and resubmission of invoices with comments and correction tracking.

Invoice Distribution

Upon successful approval, the system shall automatically distribute the invoices to the respective buyers through multiple channels.

Capabilities include:

- Automated email delivery of invoices in predefined formats.
- SMS notifications informing buyers about invoice availability.
- Buyer portal access for secure viewing and downloading of invoices.
- Tracking of invoice access history including download and viewing logs for audit and compliance purposes.

This integrated workflow shall ensure secure, traceable, and timely dissemination of invoices to all stakeholders.


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H. Payment Security and Payment Tracking

The system shall support monitoring and management of payment security mechanisms, payment status, payment reconciliation, and related contractual adjustments. The system shall also support integration with SJVN's financial/ERP systems, such as SAP or equivalent, for data exchange and reconciliation, subject to defined interfaces, access approvals, and technical feasibility.

Scope includes:

- Tracking and management of payment security mechanisms including corpus fund, payment security fund, Letter of Credit (LC), bank guarantees, or any other applicable security mechanism as defined under the relevant PPA/PSA.
- Recording and monitoring of LC details including LC amount, issuing bank, beneficiary, validity period, available limit, utilization, shortfall, expiry, renewal status, invocation status, drawdown details, claim timelines, and associated supporting documents.
- Tracking of payments received from buying entities and payments made to developers, including payment date, amount, invoice reference, transaction reference, payment mode, deductions, adjustments, and settlement status.
- Support for payment reconciliation with invoices, outstanding dues, receivables, payables, partial payments, disputed amounts, undisputed amounts, and settlement records.
- Segregation and tracking of disputed and undisputed billing amounts, including settlement status and adjustment entries after dispute resolution.
- Management of partial payments, part-capacity commissioning-related payments, deductions, refunds, recoveries, and other applicable payment adjustments.
- Configurable rule-based calculation of payment due dates, applicable rebates, Late Payment Surcharge (LPS), interest, penalties, and other contractual or regulatory adjustments.
- Monitoring of relevant performance parameters such as CUF, availability, peak availability, or other contract-specific parameters, and computation of associated penalties or adjustments as applicable.
- Alerts and notifications for payment due dates, overdue payments, LC expiry, LC shortfall, security mechanism breach, pending claims, delayed settlements, and reconciliation exceptions.
- Maintenance of payment history, claim history, LC/security mechanism history, adjustment history, reconciliation records, and audit trail.
- Support for monthly, quarterly, annual, or other periodic reconciliation of payment records, security mechanisms, rebates, LPS, penalties, deductions, and settlement adjustments, as applicable.

I. Settlement and Financial Tracking

The system shall support computation and tracking of settlements between SJVN, developers, and buying entities.

Scope includes:

- Computation of payable and receivable amounts based on billing and adjustments

- Handling of partial payments, deductions, and settlement revisions
- Maintenance of transaction-wise financial records
- Tracking of outstanding dues and settlement status

J. Reconciliation

The system shall support periodic (Yearly, Quarterly and Monthly) reconciliation of energy and financial data

Scope includes:

- Validation of energy data against billing records
- Validation of Performance criteria (CUF, Peaking availability etc.)
- Validation of Payment Records, Rebates, LPS
- Validation of Monthly, Quarterly and Annual Penalty as per PSA & PPA conditions.
- Comparison of developer invoices with system-generated values
- Identification and reporting of discrepancies
- Provision for assisted reconciliation through rule-based and pattern-based checks
- Support for periodic reconciliation on monthly, quarterly, and annual basis
- Generation of reconciliation statements for joint validation between stakeholders

K. Dispute Management

The system shall provide a structured and transparent mechanism for managing invoice-related disputes raised by buyers or sellers. The module shall allow stakeholders to review invoices, raise discrepancies, and track the dispute resolution process through defined workflows.

The system shall support the following capabilities:

Invoice Review and Dispute Initiation

Buyers and sellers shall be able to review invoices through their respective portals and raise disputes if discrepancies are identified.

Features include:

- Secure invoice viewing and download functionality through the user portal.
- Online submission of Bill Dispute Notices by buyers or sellers.
- Provision to submit comments and upload supporting documents while raising disputes.
- Recording of dispute details including invoice reference, issue description, and supporting evidence.

Dispute Processing Workflow

The system shall implement a structured and time-bound workflow for dispute handling.

Capabilities include:

- Configurable dispute resolution workflows involving multiple stakeholders.
- Generation of disagreement notices where applicable.
- Status tracking of disputes throughout the lifecycle (Submitted, Under Review, Resolved, Closed).

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- Automatic notifications and alerts to concerned stakeholders at each stage of the dispute process.

Financial and Contractual Compliance Handling

The system shall support contractual provisions related to dispute resolution including:

- Tracking of mandatory undisputed payment requirements
- Automatic revision of invoices after resolution of disputes.
- Calculation and adjustment of applicable interest or Late Payment Surcharge (LPS) on refunds or recoveries arising from dispute resolution.

L. Notification and Alert System

The system shall provide automated notification mechanisms.

Scope includes:

- Multi-level approval workflows for billing, settlement, and other processes
- Alerts for key events such as invoice generation, Invoice Approval, Dispute Submission, Dispute Resolution, payment due dates, delays, LC expiry, and reconciliation exceptions
- Notification through email, SMS and system-based alerts

M. Dashboard, Monitoring and Master Configuration

The REIA module shall provide operational, financial, compliance, and master data dashboards for authorized users. The module shall also provide required data points and MIS outputs to the Management Dashboard and Consolidated MIS layer for senior management review.

Scope includes:

- Dashboard visibility for key operational indicators, including total active contracts, contracted capacity, energy supplied versus contracted energy, billed energy, pending invoice approvals, pending disputes, settlement status, reconciliation exceptions, and compliance indicators.
- Dashboard visibility for key financial indicators, including invoices raised, total invoices generated, receivables, payables, outstanding amounts, buyer receivables, seller payable amounts, payments received, payments disbursed to developers, overdue payments, and billing status.
- User-specific dashboards for SJVN administrators, operational users, buyers, sellers, finance users, and other authorized users, based on role-based access rights.
- Reports on contracts, billing, energy data, invoices, payments, financials, disputes, reconciliation, settlement status, and compliance indicators.
- Export of reports and dashboards in standard formats such as Excel, PDF, CSV, or equivalent.
- Scheduled report generation and automated report distribution to authorized users, wherever required.
- Configurable master data modules to be managed by authorized users, including entity master, contract master, project master, bank master, regulatory parameter master, billing parameter master, and other relevant configuration masters.

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- Entity master shall include SJVN, buyers, sellers, RE generators, DISCOMs, and other relevant entities.
- Contract master shall include PPA/PSA details, contract capacity, tariff, tenure, billing rules, payment terms, and security mechanisms.
- Project master shall include project ID, project name, capacity, technology type, delivery point, commissioning details, and other project-specific parameters.
- Bank master shall include bank account details of relevant entities, wherever required for billing, payment tracking, and reconciliation.
- Regulatory and billing parameter masters shall include LPS rate, rebate slabs, penalty rules, tax parameters, payment due date rules, and other applicable contractual/regulatory parameters.
- Role-based management, approval workflow, version control, and audit tracking of master data and configuration changes.

N. User Management and Access Control

The system shall provide user management capabilities.

Capabilities include:

- Role Based Access Control (RBAC)
- User provisioning and deactivation
- Password and authentication management
- User activity tracking

3B. Scope of Work – Power Trading Management System

The selected bidder shall design, develop, implement, test, deploy, and support a Power Trading Management System for SJVN Limited to digitize, automate, and streamline its power trading operations. The system shall support exchange trading, bilateral transaction management, client and counterparty management, bid management, scheduling, open access tracking, billing, settlement, reconciliation, energy accounting, accounting support, payment tracking, compliance reporting, dashboards, MIS reporting, and integration with relevant internal and external systems.

The Power Trading Management System shall be implemented as an independent business module with separate workflows, user roles, data structures, configurations, and reporting requirements from the REIA Billing, Contract and Settlement Management System. However, both modules shall support common management-level reporting and consolidated dashboards, wherever required by SJVN.

I. Exchange Trading:

- Automatic generation, validation, approval, and submission of bids for all Power Exchange products through an integrated workflow and approval mechanism for SJVN and its clients.
- Support bidding operations for all Power Exchange products including but not limited to:
 - Day Ahead Market (DAM)
 - High Price Day Ahead Market (HPDAM)
 - Term Ahead Market (TAM)
 - Green Day Ahead Market (GDAM)

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- Real Time Market (RTM)
- Green Term Ahead Market (GTAM)
- Renewable Energy Certificate (REC)
- Energy Saving Certificate (ESCert)
- Renewable Purchase Obligation (RPO)
- Support integration or data exchange with Power Exchanges including IEX, PXIL, HPX, or any other exchange/platform as required by SJVN, subject to availability of APIs, SFTP, file-based interfaces, access credentials, and approvals from the respective exchange/platform.
- Creation and management of clients, generators, DISCOMs, traders, sellers, buyers, and other counterparties, along with user access control, role management, client onboarding, document management, PPA management, NOC management, and tracking of document validity wherever applicable.
- The system shall support role-based access control and maker-checker workflow for critical activities such as bid submission, billing, settlement, reconciliation, configuration changes, and report generation.
- Provision for bid placement through manual entry, Excel-based upload, bulk upload, copy-paste from Excel/CSV, and copy from historical bids. The system shall also provide bid edit, delete, and cancellation facility before applicable exchange cut-off timelines.
- Option to configure bid placement flow for OCF / carry-forward selection across market segments such as DAM to RTM, GDAM to DAM to RTM, and GDAM to RTM, with configurable premium/discount settings and carry-forward of uncleared quantity wherever applicable.
- Price forecasting for Day Ahead Market, Green Day Ahead Market, Real Time Market, and other market products as mutually agreed with SJVN. The final forecasting approach, data sources, model complexity, accuracy expectations, output formats, and refresh frequency shall be finalized during the requirement gathering and design phase.
- Where direct integration with exchange/platform systems is unavailable or temporarily not functioning, the system shall support manual upload/download through agreed templates or exchange-specific formats.

II. Bilateral Transaction Management:

The key functionalities of the Bilateral Transaction Management module shall include, but not be limited to:

- Submission of bilateral schedules by clients/generators.
- LOI and Contract Management.
- Schedule preparation, approval, revision, cancellation, and status tracking.
- Open Access approval tracking and status monitoring.
- Real-time or near real-time schedule monitoring against each application/approval, wherever feasible, along with alerts for application approval, rejection, partial approval, schedule revisions, and deviation between approved quantum and scheduled quantum.
- Calculation and tracking of wheeling charges, transmission charges, open access charges, applicable losses up to the delivery point, tariff components, and other applicable bilateral transaction charges as per approved business rules and available input data.
- Integration or data exchange support with SLDC/RLDC/NOAR scheduling and application processes wherever applicable and feasible, subject to availability of interfaces, access, approvals, and support from respective systems.

III. Trading Billing, Settlement and Accounting

The system shall support generation of bills as per inputs provided by the user, including trading margin bills only, power supply bills only, and combined power supply plus trading margin bills, with or without applicable taxes/GST as per configuration.

The system shall also support energy accounting and reconciliation, bilateral billing, exchange billing, payment tracking, settlement tracking, account ledger management, client ledger management, account statements, and financial reporting. The system shall also support integration readiness with ERP/accounting systems such as SAP/ERP for billing, settlement, payment, accounting, ledger, and reconciliation data exchange, subject to defined interfaces and approvals.

Provision for manual payment entry and reconciliation, along with automatic bank integration and a virtual account-based collection mechanism, wherever applicable. Digital signature and e-invoicing support shall also be provided, wherever applicable and as required by SJVN..

IV. Dashboard, MIS and Market Analytics Reporting Facility for SJVN

The system shall provide dashboards and MIS reports for SJVN trading users, finance users, management users, and other authorized stakeholders.

- Market rates and analytics
- Financial and profitability reports
- Management dashboard covering snapshots of exchange trading, bilateral transactions, billing, settlement, reconciliation, compliance, and key financial indicators.

V. Regulatory, statutory, operational, audit, and internal MIS reports, including Form-4 reports for REIA and power trading activities, wherever applicable.

VI. Updation/upgradation of software from time to time in line with statutory, regulatory, exchange, market, technology, and business requirements, subject to the agreed change management process and contract provisions.

VII. The system shall maintain user activity logs, transaction logs, approval logs, configuration change logs, and audit trail for key trading, billing, settlement, reconciliation, and reporting activities.

3C. Management Dashboard and Consolidated MIS Layer

The proposed platform shall provide a Management Dashboard and Consolidated MIS layer to enable SJVN leadership and authorized users to monitor operational, commercial, financial, reconciliation, compliance, and profitability performance across both business streams:

1. REIA Billing, Contract and Settlement Management System
2. Power Trading Management System

The dashboard layer shall provide both module-wise independent views and consolidated management-level views, while maintaining appropriate segregation of data, roles, workflows, and access controls for each business stream.



Scope includes:

- Separate dashboards for REIA billing operations and power trading operations, covering module-specific operational, commercial, financial, reconciliation, compliance, and reporting parameters.
- Consolidated executive dashboard for senior management to view overall portfolio position, REIA billing status, power trading status, contracts, schedules, bid status, obligations, invoices, payments, receivables, payables, outstanding dues, settlements, disputes, reconciliation exceptions, compliance status, trading margin, profitability, and overall financial performance.
- Power trading dashboards covering exchange trading, bilateral transactions, market rates and analytics, bid placed versus cleared position, client-wise transaction summary, exchange-wise and product-wise transaction summary, account ledger, payment status, trading margin, profitability, and regulatory reports.
- REIA dashboards covering contracted capacity, scheduled energy, billed energy, invoices raised, receivables, payables, payment security status, settlement status, dispute status, reconciliation exceptions, and compliance indicators.
- Module-wise and consolidated MIS reports covering contracts, trading transactions, invoices, billing status, payment status, settlement status, reconciliation status, disputes, compliance submissions, financial exposure, and profitability.
- Drill-down capability from consolidated dashboard to module-wise, entity-wise, client-wise, contract-wise, exchange-wise, product-wise, invoice-wise, schedule-wise, bid-wise, or transaction-wise details, based on user access rights.
- Visibility of key management indicators such as total contracted capacity, scheduled energy, billed energy, invoices raised, payments received, payments made, outstanding amounts, overdue receivables, trading margin, profitability, pending approvals, open disputes, reconciliation exceptions, and compliance status.
- Role-based access to dashboards and reports for SJVN management, business users, finance users, REIA users, trading users, administrators, and other authorized users.
- Configurable report templates, filters, date ranges, parameters, user-specific views, and dashboard widgets.
- Export of reports and dashboards in standard formats such as Excel, PDF, CSV, or equivalent.
- Scheduled generation and automated distribution of MIS reports to authorized users, wherever required.
- Maintenance of audit trail for report generation, access, download, and distribution, wherever applicable.
- Provision to add or modify KPIs, dashboards, MIS reports, and report formats in future based on business, regulatory, operational, or management requirements.

3D. Real Time Integration with Existing ERP System (SAP)

- Real-time integration with the existing ERP (SAP) system for seamless recording, processing, and monitoring of all financial transactions.



4. Technical, Security, Infrastructure and Implementation Requirements

The proposed platform shall be designed, developed, and implemented using modern, secure, scalable, and interoperable technologies, aligned with industry best practices, to ensure long-term sustainability, flexibility, and ease of integration with existing and future systems.

The technical requirements shall apply to both proposed systems/modules, namely the REIA Billing, Contract and Settlement Management System and the Power Trading Management System, along with the Management Dashboard and Consolidated MIS layer. The platform shall support appropriate logical separation of business processes, user roles, workflows, data structures, configurations, and reporting for each module, while enabling common platform services such as authentication, notifications, audit logs, integrations, and consolidated dashboards.

4.1 Architecture and Application Design

The proposed solution shall be designed using a modern, scalable, secure, configurable, and maintainable architecture capable of supporting SJVN's growing renewable energy portfolio, power trading operations, increasing transaction volumes, future integrations, and evolving regulatory/business requirements.

The solution architecture shall support the following:

Application Architecture

- Follow a modular, layered, service-oriented, component-based, or microservices-based architecture approach to ensure scalability, maintainability, security, and independent extensibility of functional modules.
- Support logical separation of the REIA Billing, Contract and Settlement Management module and the Power Trading Management module, including separate business workflows, data structures/schemas, access controls, configuration parameters, reports, dashboards, and approval hierarchies.
- Enable consolidated reporting and management dashboards across both modules through a common Management Dashboard and MIS layer, without compromising module-wise data segregation and role-based access controls.
- Support common platform services across modules such as authentication, authorization, notifications, audit logging, configuration management, document management, reporting services, integration services, and monitoring services, wherever applicable.
- Support separation of presentation layer, business logic layer, workflow/rule engine layer, integration layer, reporting layer, and data layer to facilitate maintainability, scalability, security, and future enhancements.
- Enable deployment using modern application frameworks and open standards. Examples may include Java Spring Boot, .NET Core, Node.js, Python-based frameworks, or equivalent enterprise-grade technologies.



- Support configurable and extensible design to accommodate new contract types, market products, exchange products, regulatory requirements, reporting requirements, and future business processes without requiring major redevelopment.

User Interface Layer

- Provide a responsive web-based user interface compatible with major browsers such as Chrome, Edge, Firefox, and other standards-compliant browsers.
- Support responsive design principles to ensure accessibility across desktops, laptops, tablets, and other supported devices.
- Utilize modern frontend frameworks such as Angular, React, Vue.js, or equivalent technologies.
- Provide role-based user interfaces, dashboards, menus, workflows, forms, and reports for SJVN REIA users, Trading users, Finance users, Management users, Administrators, Buyers, Sellers, Generators, DISCOMs, Traders, external trading clients, and other authorized users, as applicable.
- Support user-friendly screens for data entry, bulk upload, approval workflows, invoice processing, bid management, schedule management, reconciliation, dashboards, and reporting.
- Provide client-facing portal capabilities for authorized external users to view and manage relevant transactions, bids, schedules, invoices, obligations, payments, account statements, documents, and reports based on access rights.

Business Rules and Workflow Engine

- Support configurable workflow and rule management capabilities to accommodate changing business requirements without requiring extensive application redevelopment.
- Enable configurable approval hierarchies, maker-checker controls, validation rules, billing logic, settlement calculations, reconciliation rules, penalty computations, document approvals, notification workflows, and exception handling.
- Support REIA-related workflows including stakeholder onboarding, contract management, energy data validation, invoice generation, invoice approval, payment tracking, payment security management, reconciliation, dispute management, and settlement tracking.
- Support power trading workflows including client onboarding, NOC/PPA validity checks, bid creation, bid validation, bid approval, bid edit/delete/cancellation, No Bid submission, bid submission tracking, exchange obligation processing, bilateral schedule submission, schedule approval/revision/cancellation, NOAR/open access tracking, billing, settlement, payment reconciliation, client ledger, and compliance reporting.
- Support configurable bid validation rules including pre-payment balance validation, margin requirement validation, NOC/PPA validity validation, quantum validation,

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time-block validation, product eligibility validation, and exchange-specific validation rules.

- Support configurable OCF/carry-forward bid logic across market segments such as DAM to RTM, GDAM to DAM to RTM, GDAM to RTM, or other combinations as may be required by SJVN, including configurable premium/discount parameters and audit trail of carried-forward bids.
- Be configurable to accommodate regulatory changes, policy updates, tariff revisions, contractual amendments, new RE product structures, power market product changes, exchange rule changes, settlement cycle changes, tax/GST changes, reporting format changes, and evolving business requirements through parameterized business rules and configuration-driven workflows.

Integration Architecture

- Support integration through standard RESTful APIs, SOAP services where required, secure file-based exchange mechanisms, SFTP, scheduled data uploads/downloads, database views/interfaces, message queues, and other industry-standard integration approaches.
- Support standard data interchange formats such as JSON, XML, CSV, Excel, PDF, and other structured formats as applicable.
- Enable integration with ERP/finance/accounting systems, energy data sources, power exchanges, NOAR/NLDC/RLDC/SLDC-related systems, banking systems, virtual account collection mechanisms, digital signature/e-invoicing systems, email gateways, SMS gateways, payment systems, and other external applications as identified during implementation.
- Support automatic import and processing of exchange-related files such as obligation reports, scheduling files, settlement files, invoices, rate sheets, account statements, and other relevant data files, subject to availability of APIs/SFTP/interfaces.
- Provide fallback mechanisms such as manual upload/download of exchange-specific prescribed formats in case of API/SFTP failure, downtime, or connectivity issues.
- Support integration monitoring, error handling, retry mechanisms, exception reporting, and reconciliation logs for all critical integrations.

Data Architecture and Management

- Support logical data segregation between the REIA Billing module and Power Trading module, with common reference data and consolidated reporting views wherever required.
- Maintain complete and structured data for stakeholders, clients, generators, DISCOMs, traders, contracts, PPAs, PSAs, NOCs, exchange products, bids, schedules, obligations, invoices, payments, ledgers, settlements, reconciliations, disputes, compliance records, documents, and audit logs.

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- Support bulk upload, template-based data entry, validation, data correction, version management, and audit trail for critical master and transaction data.
- Support real-time or near real-time synchronization between the Trading Portal, exchange APIs/SFTP data sources, internal database systems, dashboards, and reporting layer, subject to availability and performance of external interfaces.
- Support data archival, retrieval, backup, retention, and purging mechanisms in line with SJVN's data governance, audit, and compliance requirements.

Scalability and Performance

- Support horizontal and/or vertical scaling mechanisms to accommodate increasing user loads, transaction volumes, contract data, invoice processing, billing runs, trading transactions, bid records, schedule data, exchange files, reconciliation records, compliance records, forecasting data, and reporting workloads.
- Support deployment architectures capable of load balancing, failover, clustering, and high availability, as per the approved deployment architecture.
- Be capable of handling concurrent users, high-volume invoice processing, exchange trading workflows, bilateral trading workflows, bid submission windows, bulk uploads, scheduled jobs, reconciliation runs, large-scale contract and transaction data, and management dashboards without performance degradation.
- Support optimized processing for bulk bid uploads, exchange file processing, billing computations, invoice generation, payment reconciliation, client ledger updates, dashboard refreshes, and report generation.

Messaging, Scheduling and Background Processing

- Support asynchronous processing mechanisms for handling long-running operations, notifications, integrations, bulk uploads, batch processing, exchange file imports, bid processing, billing runs, scheduled reconciliation, report generation, data synchronization, and background jobs.
- Support scheduler-based processing for periodic tasks such as energy data ingestion, exchange data import, invoice generation, obligation processing, reconciliation, compliance report generation, dashboard refresh, and notification dispatch.
- Examples may include message queues, event-driven processing frameworks, scheduler services, batch processors, background workers, or equivalent technologies.

Analytics and Forecasting Readiness

- Support dashboarding, analytics, and reporting capabilities for operational, commercial, financial, compliance, market, and management-level insights.
- Support price forecasting and market analytics capabilities for Day Ahead Market, Green Day Ahead Market, Real Time Market, and other products as mutually agreed with SJVN.

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- The architecture shall be capable of supporting historical data analysis, market rate analytics, forecasted versus actual comparison, trend analysis, and integration with analytical models, subject to finalization of data sources, modelling approach, refresh frequency, and accuracy expectations during the requirement and design phase.

Auditability and Traceability

- Maintain complete traceability of user actions, workflow approvals, billing calculations, bid creation, bid modification, bid cancellation, bid approval, bid submission, No Bid entries, schedule changes, contract changes, NOC/PPA updates, configuration changes, invoice actions, payment entries, reconciliation outcomes, settlement activities, report generation, and system-generated transactions.
- Support centralized logging and audit mechanisms for monitoring, troubleshooting, compliance, security review, internal audit, regulatory review, and management reporting.
- Maintain version history and audit trail for critical master data, contracts, PPAs/PSAs, NOCs, billing configurations, trading margin configurations, workflow configurations, reports, and system parameters.

4.2 Security Requirements

The solution shall incorporate appropriate security controls to ensure confidentiality, integrity, availability, and protection of application data, user access, business transactions, and system resources. The security design shall follow industry best practices and applicable Government of India cybersecurity guidelines.

The solution shall support:

- Role-Based Access Control (RBAC) with configurable roles, permissions, and access rights for SJVN REIA users, Trading users, Finance/Accounting users, Management users, Administrators, Buyers, Sellers, Generators, DISCOMs, Traders, external trading clients, and other authorized stakeholders.
- Secure authentication and authorization mechanisms, including support for password policies, session timeout, account lockout, access control rules, and integration with enterprise identity systems such as LDAP, Active Directory, SSO, SAML, OAuth, OpenID Connect, or equivalent standards, where required.
- Secure API and application access using standard mechanisms such as token-based authentication, API keys, JWT, OAuth, digital certificates, IP whitelisting / IP-based monitoring, or equivalent methods, wherever applicable.
- Protection against common application vulnerabilities in line with OWASP guidelines, secure coding practices, input validation, error handling, and security testing practices.
- Encryption of data in transit using industry-standard secure communication protocols such as HTTPS/TLS and encryption of sensitive data at rest, wherever applicable.
- Comprehensive audit logs for user activities, login attempts, role/permission changes, configuration changes, approvals, bid creation, bid modification, bid cancellation, bid submission, schedule revision, invoice actions, billing actions, payment entries,

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contract changes, NOC/PPA updates, reconciliation activities, settlement operations, report access, and other key business transactions.

- Security monitoring, application logging, vulnerability management, patch management, access review, and compliance with applicable MeitY, CERT-In, and Government of India cybersecurity guidelines, wherever applicable.
- Appropriate controls for secure file upload/download, bulk uploads, exchange file processing, document storage, and integration data exchange, including validation, access restriction, and auditability.
- Secure management of integrations with ERP/accounting systems, power exchanges, NOAR/NLDC/RLDC/SLDC-related systems, banking/virtual account systems, digital signature/e-invoicing systems, email/SMS gateways, and other external systems, wherever applicable.
- Periodic security review and remediation support during implementation and O&M, as mutually agreed with SJVN.

4.3 Infrastructure and Deployment Requirements

The solution shall be deployed on a secure, scalable, and compliant cloud environment as finalized by SJVN. The deployment architecture shall follow industry best practices for availability, performance, security, monitoring, backup, recovery, and business continuity.

The solution shall support:

- Deployment on a MeitY-empanelled cloud service provider or any other cloud environment approved by SJVN, in compliance with applicable Government of India and SJVN IT/security requirements.
- Cloud infrastructure sizing based on expected users, concurrent access, transaction volumes, data storage, integration load, REIA billing workloads, power trading workloads, bid volumes, exchange transactions, schedule data, invoice volumes, reconciliation records, market data, forecasting workloads, dashboard/reporting workloads, and future scalability requirements.
- Scalable deployment architecture to support increasing users, contracts, invoices, energy data, trading bids, schedules, exchange transactions, reconciliation records, compliance records, reports, and dashboards.
- High availability and resilience through appropriate cloud-native mechanisms such as load balancing, auto-scaling, redundancy, failover, and resource monitoring, wherever applicable.
- Secure network architecture including controlled access, firewall/security group configuration, secure connectivity, and segregation of application, database, integration, reporting, and external access components.
- Environment separation for development, testing/UAT, staging/pre-production, and production, as required, with controlled access and defined release movement procedures.
- Secure storage, backup, archival, and recovery mechanisms for application data, trading data, billing data, documents, logs, reports, configuration data, and audit records.
- Scalable background processing for exchange file import, bid processing, billing runs, reconciliation jobs, notification processing, report generation, data synchronization, and other scheduled/batch activities.
- Monitoring of infrastructure performance, application availability, integration health, exchange/API/SFTP job status, background job status, system logs, resource utilization, and alerts for failures, exceptions, or threshold breaches.

- Disaster recovery and business continuity provisions as per SJVN's requirements.
- Deployment automation and release management practices to support controlled, auditable, and repeatable deployments.

4.4 Integration and Data Management

The solution shall support secure, reliable, and scalable integration with external systems and data sources required for REIA billing, energy accounting, power trading, exchange operations, scheduling, reconciliation, settlement, reporting, compliance, and operational monitoring. The integration architecture shall be based on open standards and industry best practices to ensure interoperability, security, traceability, and future extensibility.

The solution shall support:

- Integration with relevant external data sources and systems, including but not limited to RLDC, SLDC, REA/SEA data sources, JMR data, RE generator/seller systems, buyer/DISCOM systems, trading clients, power exchanges such as IEX/PXIL/HPX, NOAR/NLDC/RLDC/SLDC-related systems, SJVN ERP/finance/accounting systems, banking systems, virtual account collection mechanisms, digital signature/e-invoicing systems, email/SMS gateways, and other systems as identified during implementation.
- Data exchange through standard integration mechanisms such as REST APIs, SOAP APIs, SFTP, secure file transfer, scheduled data uploads/downloads, database views/interfaces, message queues, or other mutually agreed mechanisms based on source system capability.
- Support for commonly used data exchange formats such as JSON, XML, CSV, Excel, PDF, and other structured formats as applicable for energy accounts, schedules, bids, exchange results, obligations, invoices, rate sheets, account statements, payment records, reconciliation statements, compliance reports, and MIS reports.
- Secure data exchange using appropriate protocols and controls such as HTTPS, SFTP, VPN/private connectivity, API authentication, encryption, digital certificates, token-based access, IP-based controls, or equivalent security mechanisms, wherever applicable.
- Automated or semi-automated ingestion of energy data, schedule data, bid data, exchange data, obligation data, settlement data, invoice data, payment data, contract data, PPA data, NOC data, client data, master data, market rate data, forecasting data, ledger data, and compliance data.
- Automatic import of obligation, scheduling, settlement, invoice, rate sheet, account statement, and other exchange/client files through API/SFTP or other approved mechanisms, subject to availability of source interfaces.
- Provision for manual upload/download through predefined templates where direct system integration is not available, not feasible, or temporarily not functioning.
- Data mapping, transformation, validation, exception handling, and reconciliation between source systems and the proposed application.
- Validation checks to identify incomplete, duplicate, inconsistent, erroneous, or unmatched records during data upload, ingestion, and reconciliation.
- Maintenance of complete transaction history, data upload logs, integration logs, error logs, retry logs, reconciliation records, and audit trails.
- Data backup, archival, retrieval, retention, and recovery mechanisms in line with SJVN's requirements and applicable regulatory or compliance obligations.







- Scalability to support future integrations with additional market platforms, regulatory systems, forecasting tools, banking/payment systems, power exchanges, or other third-party applications as required by SJVN.

4.5 Deployment, Training and Change Management

The bidder shall be responsible for deployment, configuration, rollout, user training, handholding, and change management support for successful adoption of the REIA Billing, Contract and Settlement Management System, Power Trading Management System, and Management Dashboard / Consolidated MIS layer. The implementation shall follow a structured project management approach, preferably based on Agile / iterative delivery methodology, with defined milestones, sprint/release planning, periodic reviews, and formal sign-offs.

The bidder shall be responsible for:

- Preparing and submitting a detailed deployment and implementation plan covering project milestones, sprint/release plan, environment readiness, configuration, data migration or data setup, testing, integration testing, cutover, go-live, stabilization activities, dependencies, and module-wise rollout for both proposed systems and the consolidated dashboard layer.
- Deploying and configuring the application on the cloud infrastructure finalized by SJVN, including required application components, databases, integrations, access controls, workflows, system parameters, business rules, reports, dashboards, and monitoring components.
- Setting up separate environments for development, testing/UAT, staging/pre-production, and production, as required, with defined access controls, configuration management, and release movement procedures.
- Undertaking data migration or initial data setup, if applicable, including data extraction, cleansing, transformation, upload, validation, reconciliation, and exception handling from existing sources/templates for both modules.
- Configuring business rules, workflows, user roles, approval hierarchies, notifications, reports, dashboards, billing rules, trading rules, reconciliation rules, settlement parameters, NOC/PPA validations, OCF/carry-forward configuration, and other parameters as per approved requirements.
- Supporting integration testing with external systems such as ERP/finance/accounting systems, energy data sources, power exchanges, NOAR/NLDC/RLDC/SLDC-related systems, banking/virtual account systems, digital signature/e-invoicing systems, email/SMS gateways, and other systems identified during implementation.
- Preparing and executing go-live plans covering production deployment, user readiness, data readiness, access provisioning, integration readiness, report/dashboard readiness, backup readiness, rollback approach, module-wise rollout, and post-go-live support.
- Ensuring deployment readiness in line with applicable security and compliance requirements, including CERT-In readiness, MeitY guidelines, and other relevant Government of India cybersecurity advisories, wherever applicable.
- Providing user training for SJVN REIA users, Trading users, Finance/Accounting users, Management users, Administrators, Buyers, Sellers, Generators, DISCOMs, Traders, external clients, and other authorized users through structured training sessions, user manuals, administrator guides, quick reference guides, and reference materials.

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- Ensuring that training and documentation cover REIA billing operations, exchange trading workflows, bilateral transaction workflows, bid management, bid edit/delete/cancellation, No Bid process, OCF/carry-forward process, NOC/PPA management, billing/settlement, reconciliation, payment entry, client ledger, dashboards, reports, compliance reporting, and fallback procedures for manual exchange upload/download.
- Providing handholding and stabilization support during the initial rollout period to address user queries, operational issues, configuration changes, data issues, integration issues, and minor refinements.
- Supporting change management activities including stakeholder orientation, process familiarization, communication support, user adoption support, and transition from manual/existing processes to the new system.
- Maintaining relevant project, deployment, technical, training, and change management documentation, including but not limited to System Requirement Specification (SRS), Functional Design Document (FDD), Technical Design Document (TDD), System Architecture Document, API / Integration Document, Database Design Document, Security Design Document, Test Plan and Test Reports, Deployment Guide, User Manual, Administrator Manual, Training Material, Release Notes, and Go-Live Sign-off documentation.

5. Operations & Maintenance (O&M)

The selected bidder shall provide post-implementation Operations & Maintenance (O&M) support for the complete platform for the duration specified in the contract for a period of three (3) years and further extendable to two (2) years on same terms & conditions. The O&M support shall cover the REIA Billing, Contract and Settlement Management System, the Power Trading Management System, common platform services, integrations, dashboards, reports, and related application components.

The O&M phase shall commence after successful go-live and stabilization of the system, as approved by SJVN. During the O&M period, the bidder shall ensure stable, secure, reliable, and efficient functioning of the platform in accordance with the agreed service levels.

5.1 Scope of O&M

The scope of O&M shall include, but not be limited to:

- Provision of **2 nos. dedicated manpower** to carry out scope of mentioned at 3.0. The manpower should be familiar with carrying out the commercial & IT activities.
- Application support for the REIA Billing module, Power Trading module, Management Dashboard / Consolidated MIS layer, common platform services, and related application components.
- Monitoring and maintenance of application performance, availability, reliability, response time, integration health, background jobs, scheduled processes, and critical business workflows.
- Resolution of application issues, defects, bugs, functional errors, integration issues, data issues, configuration-related issues, and user-reported incidents within agreed timelines.
- Support for common platform services including user management, role configuration, access control, workflows, notifications, dashboards, reports, audit logs, configuration management, and document management.

- Monitoring and support of integrations with ERP/finance/accounting systems, energy data sources, exchange API/SFTP jobs, exchange/market platforms, NOAR/NLDC/RLDC/SLDC-related systems, banking/virtual account systems, digital signature/e-invoicing systems, email/SMS gateways, and other connected systems, wherever applicable.
- Implementation of approved updates, patches, corrective fixes, minor enhancements, configuration changes, and release updates.
- Support for changes arising from regulatory requirements, policy updates, tariff changes, contractual amendments, statutory changes, tax/accounting changes, market rule changes, exchange product/format changes, trading business process changes, or other approved business process modifications.
- Routine system administration support including user access support, role updates, logs review, backup monitoring, archival support, recovery assistance, configuration review, and issue tracking.
- Application of security patches and corrective fixes in line with applicable security practices, vulnerability remediation requirements, and SJVN's IT/security requirements.
- Support for price forecasting and market analytics configuration, monitoring, and updates as per agreed scope, available data, finalized methodology, and approved requirements.
- Support for issue resolution related to REIA billing workflows, invoice processing, payment tracking, dispute handling, reconciliation, power trading workflows, bid management, OCF/carry-forward bidding, NOC/PPA validity checks, obligation import, settlement file processing, payment reconciliation, client ledger, dashboards, reports, and Form-4 reporting.
- Maintenance of issue logs, change logs, release notes, configuration records, integration records, audit logs, SLA records, and system performance reports.
- Periodic review meetings with SJVN to review open issues, SLA compliance, system performance, integration health, risks, dependencies, change requests, and upcoming business or regulatory requirements.

5.2 Support Window

The bidder shall provide application support during SJVN's working hours on all working days. Critical issues impacting system availability, billing operations, power trading operations, bid submission, exchange-related processing, payment processing, reconciliation, settlement, or management reporting shall be supported on priority basis, including beyond working hours where required and agreed with SJVN.

Support requirements for critical bidding windows, exchange operations, billing cycles, settlement activities, regulatory deadlines, or other time-sensitive business processes shall be handled on priority based on severity and business impact.

5.3 Issue Severity Classification

Issues shall be classified based on severity and business impact as defined below:

Severity Level	Description	Indicative Examples
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Severity 1 – Critical	Complete system outage or failure of critical business functionality causing stoppage of key operations, where no immediate workaround is available	Application unavailable; REIA billing module not accessible; Power Trading module unavailable during critical bidding/trading window; inability to generate or approve critical invoices; major production database access issue; major security incident; exchange integration failure preventing bid submission or obligation import during critical window; system failure impacting billing, settlement, or trading operations at platform level
Severity 2 – High	Major functionality impacted with significant business or operational impact, but business operations can continue through an available workaround	Invoice generation failure for selected contracts or clients; bid submission workflow failure for selected products/clients; OCF/carry-forward bid logic failure; incorrect bid validation affecting selected clients/products; failure in obligation, scheduling, settlement, or exchange file import with workaround available; integration failure with ERP/accounting, exchange, NOAR/SLDC/RLDC, banking, or data source where workaround is available; payment reconciliation or client ledger issue affecting selected clients; approval workflow failure impacting selected transactions
Severity 3 – Medium	Non-critical functionality impacted, causing limited operational impact; core business processes can continue without major disruption	Report mismatch; dashboard data mismatch or delayed refresh; user role or configuration issue; non-critical validation issue; delayed email/SMS/portal notification; issue in non-critical MIS report; minor issue in data upload template; reconciliation exception report issue where manual validation is possible
Severity 4 – Low	Minor issue, cosmetic defect, user query, clarification, small configuration change, or low-impact enhancement request	UI alignment issue; label/text correction; minor report format change; helpdesk query; minor dashboard filter change; user guidance request; small master data configuration correction; minor change in notification text or report layout

5.4 Service Level Agreement (SLA)

The bidder shall adhere to the following indicative service levels during the O&M period:

Severity Level	Response Time	Resolution / Workaround Time	Final Resolution Target
Severity 1 – Critical	Within 1 hour	Within 4 hours	Within 1 business day
Severity 2 – High	Within 2 hours	Within 8 working hours	Within 2 business days
Severity 3 – Medium	Within 1 business day	Within 3 business days	Within 5 business days
Severity 4 – Low	Within 2 business days	Within 5 business days	As mutually agreed with SJVN

Where final resolution is dependent on external systems, cloud service providers, exchanges, ERP/finance systems, data availability, third-party services, regulatory inputs, or approvals from SJVN, the bidder shall provide a suitable workaround wherever feasible and submit a resolution plan with timelines for SJVN's review.

5.5 System Availability

The bidder shall ensure that the application remains available as per agreed service levels, excluding planned downtime, scheduled maintenance windows, force majeure events, and outages attributable to cloud service provider, external systems, exchange platforms, regulatory portals, or other third-party systems beyond the bidder's control.

Parameter	Expected Service Level
Application availability	Minimum 99.0% during agreed support period
Planned maintenance	To be scheduled with prior approval/intimation to SJVN
Planned downtime communication	Minimum 48 hours prior notice, except emergency maintenance
Emergency maintenance	To be communicated to SJVN at the earliest possible time

5.6 Change Request and Enhancement Support

Any change beyond the agreed scope, including major functional enhancements, new modules, major regulatory changes, new integrations, substantial workflow changes, exchange/product changes, or major reporting changes, shall be managed through a formal Change Request (CR) process.

The CR process shall include:

- Submission of change request by SJVN.
- Impact assessment by the bidder covering effort, timeline, cost, dependencies, and risk.
- Approval by SJVN prior to implementation.
- Development, testing, deployment, and documentation of approved changes.
- Submission of release notes and version update documentation.

Minor configuration changes and minor enhancements may be handled as part of O&M, subject to effort limits and mutual agreement between SJVN and the bidder.

5.7 O&M Reporting and Review

The bidder shall submit periodic O&M reports to SJVN covering:

- Summary of issues raised, resolved, pending, and escalated.
- SLA compliance status.
- System availability and performance status.
- Integration health and failure reports, wherever applicable.
- Change requests and enhancement status.
- Security patches, updates, and releases deployed.
- Key risks, dependencies, and support observations.

Periodic review meetings shall be conducted with SJVN to monitor O&M performance, review pending issues, assess SLA compliance, and plan upcoming changes or releases.

5.8 Penalty / Service Credit

SJVN may define suitable penalty or service credit provisions in the contract for repeated SLA non-compliance, delayed resolution of critical issues, or persistent performance failures attributable to the bidder. Such penalties, if applicable, shall be linked to severity, frequency, business impact, and duration of non-compliance.

6. Deliverables

The selected bidder shall submit all deliverables required for successful design, development, testing, deployment, training, operationalization, and O&M transition of the proposed systems. The deliverables shall cover both the REIA Billing, Contract and Settlement Management System and the Power Trading Management System, along with the Management Dashboard and Consolidated MIS layer.

All deliverables shall be submitted in soft copy and, where required by SJVN, in editable formats. Deliverables shall be subject to review, comments, revision, and acceptance by SJVN.

The deliverables shall broadly include, but not be limited to, the following:

Phase / Area	Key Deliverables	Description / Coverage	Acceptance Criteria
Project Initiation and Planning	Project Plan and Governance Framework	Detailed project plan covering implementation approach, timelines, milestones, resource deployment, stakeholder coordination, communication plan, escalation matrix, risk management approach, quality assurance approach, and governance mechanism for the REIA Billing, Contract and Settlement Management System, Power Trading Management System, and Management Dashboard / Consolidated MIS layer. The project plan shall clearly identify module-wise activities, dependencies, integration activities, UAT timelines, go-live approach, and stabilization support.	Submission and approval by SJVN
Requirement Gathering and Analysis	System Requirement Specification (SRS) / Requirement Specification Document	Functional and non-functional requirements covering all approved scope areas. The SRS shall separately capture requirements for the REIA Billing, Contract and Settlement Management System, Power Trading Management System, and Management Dashboard / Consolidated MIS layer. For the Power Trading	Sign-off by SJVN after requirement validation workshops

		module, the SRS shall capture detailed requirements for exchange trading, bilateral transaction management, NOC/PPA management, client/counterparty onboarding, bid validation, OCF/carry-forward bidding, billing and settlement, digital signature/e-invoicing, payment reconciliation, client ledger, price forecasting, dashboards, reports, and integrations.	
Business Process Design	Business Process and Workflow Document	Documentation of current and proposed business processes, approval flows, user journeys, operational workflows, exception handling, and reporting processes. The document shall separately cover REIA billing operations and power trading operations, including bilateral trading, exchange trading, bid submission and approval workflows, bid edit/delete/cancellation workflows, No Bid process, OCF/carry-forward bid flows, scheduling / NOAR / open access workflows, billing, reconciliation, settlement, payment tracking, client ledger, compliance reporting, and consolidated MIS processes.	Review and approval by SJVN
Functional Design	Functional Design Document (FDD)	Module-wise functional design covering screen flows, user roles, access rights, business rules, validation rules, workflow logic, approval hierarchy, forms, reports, dashboards, alerts, notifications, and functional behavior for each module. The FDD shall include detailed design for REIA billing workflows, power trading workflows, exchange product configuration, NOC/PPA validation, bid processing rules, OCF/carry-forward logic, billing configurations, settlement logic, payment reconciliation, client ledger, price forecasting outputs, compliance reports, and consolidated dashboards.	Approval by SJVN prior to development

Technical Design	Technical Design Document (TDD) / System Design Document	Technical architecture, application components, database design, data structures/schemas, integration approach, API/interface specifications, security design, deployment architecture, common platform services, reporting layer, and consolidated dashboard architecture. The TDD shall include exchange API/SFTP integration design, ERP/accounting integration design, banking/virtual account integration design, digital signature/e-invoicing integration design, data synchronization approach, fallback/manual upload approach, price forecasting architecture, audit logging, and security controls.	Approval by SJVN / SJVN IT team
Development and Configuration	Developed and Configured Application	Development and configuration of both independent business modules and the consolidated dashboard layer as per approved design. This shall include module-specific workflows, business rules, billing logic, reconciliation logic, alerts, dashboards, reports, roles, access controls, integrations, configuration parameters, and audit trails. For the Power Trading module, development shall cover exchange trading, bilateral transaction management, client/counterparty management, NOC/PPA management, advanced bid upload, bid edit/delete/cancellation, No Bid option, OCF/carry-forward logic, billing/settlement configuration, payment reconciliation, client ledger, digital signature/e-invoicing, price forecasting, dashboards, and reports.	Demonstration and acceptance during SIT/UAT
Consolidated Dashboard and MIS Layer	Dashboard Design, MIS Report Catalogue, and Configured Dashboards	Design and configuration of module-wise and consolidated dashboards covering REIA Billing, Power Trading, financial performance, reconciliation status, outstanding dues, receivables, payables, trading margin, profitability, compliance status, operational performance, and management KPIs. Dashboard	Demonstration and approval by SJVN during UAT

		deliverables shall include market rates and analytics, bid placed versus cleared status, exchange-wise/product-wise/client-wise summaries, bilateral transaction status, Form-4 reports, client ledger reports, payment status, and consolidated management dashboard covering all modules.	
Integration Development	Integration Components and Interface Documentation	Interfaces / APIs / SFTP / file exchange mechanisms for integration with ERP/finance/accounting systems, energy data sources, RLDC/SLDC/REA/SEA/JMR data sources, power exchanges, NOAR/NLDC/RLDC/SLDC-related systems, banking/virtual account systems, digital signature/e-invoicing systems, email/SMS gateways, and other systems identified during implementation. Integration deliverables shall include interface specifications, data mapping, error handling approach, reconciliation logs, fallback/manual upload-download process, and integration testing evidence.	Successful integration testing and documentation approval
Data Migration / Data Setup	Data Migration Plan and Data Validation Report	Migration or initial setup of master data, contract data, stakeholder data, buyer/seller data, trading client data, generator/DISCOM/trader data, PPA/PSA data, NOC data, exchange product data, market/product configuration data, billing data, payment data, ledger data, historical/reference data, and other agreed data sets for both systems. The deliverable shall include data templates, validation rules, cleansing approach, upload logs, exception handling, and reconciliation reports.	Validation and sign-off by SJVN
Testing	Test Plan, Test Cases, and Test Reports	Test strategy, test scenarios, test cases, unit testing, system integration testing, regression testing, performance testing as applicable, defect logs, and closure reports. Testing shall include module-wise testing for REIA Billing and Power Trading modules,	Successful completion of testing and defect closure

		integration testing, common service testing, and validation of consolidated dashboards and MIS reports. For Power Trading, testing shall include bid processing, bid edit/delete/cancellation, No Bid, OCF/carry-forward logic, NOC/PPA validations, exchange file import, obligation/scheduling/settlement processing, billing/settlement logic, payment reconciliation, client ledger, price forecasting outputs, digital signature/e-invoicing, and <u>fallback upload/download process</u> .	
User Acceptance Testing	UAT Support and UAT Closure Report	Support to SJVN users during UAT, including test execution support, issue resolution, UAT observations, defect closure, and UAT closure report. UAT shall be conducted for REIA Billing users, Power Trading users, Finance/Accounting users, Management users, Administrators, Buyers, Sellers, external clients, and other stakeholder groups, as applicable. UAT scenarios shall cover exchange trading, bilateral trading, billing, reconciliation, settlement, reporting, dashboards, integrations, and compliance reporting.	UAT sign-off by SJVN
Deployment and Go-Live	Deployment Plan, Go-Live Plan, and Go-Live Readiness Report	Production deployment approach, release plan, access provisioning, cutover plan, rollback approach, production checklist, go-live readiness assessment, and module-wise rollout plan covering both systems, common services, integrations, and consolidated dashboard availability. Deployment planning shall include readiness of integrations, user access, master data, configuration parameters, reports, dashboards, backup, monitoring, and post-go-live support.	Go-live approval and sign-off by SJVN
Training and Knowledge Transfer	Training Materials and Training Completion Report	User manuals, administrator manuals, training presentations, quick reference guides, and training sessions for REIA Billing users, Power Trading users,	Completion of training and acceptance by SJVN

		Finance/Accounting users, Administrators, Buyers, Sellers, external clients, Management dashboard users, and other authorized users, as applicable. Training shall include REIA billing operations, exchange trading operations, bilateral transaction workflows, bid management, OCF/carry-forward process, NOC/PPA management, billing/settlement, payment reconciliation, client ledger, dashboards, reports, compliance reporting, and manual fallback procedures for exchange upload/download.	
Documentation	Complete System Documentation	Functional documentation, technical documentation, system architecture document, database design document, API/interface document, security design document, deployment guide, configuration guide, administrator guide, module-wise user manuals, reporting guide, consolidated dashboard user guide, release notes, and SOPs. Documentation shall also include exchange upload/download SOPs, bid workflow guide, billing and settlement guide, reconciliation guide, client ledger guide, price forecasting documentation, dashboard guide, digital signature/e-invoicing guide, and compliance reporting guide.	Submission and approval by SJVN
Change Management	Change Request and Release Documentation	Change request log, impact assessment, approved enhancement details, release notes, deployment records, version history, and documentation for approved changes across both systems and the consolidated dashboard layer. The change management process shall cover changes related to regulatory requirements, exchange products, market rules, billing rules, tax/GST changes, report formats, integrations, workflows, and business process changes.	Approval of change implementation by SJVN

Compliance and Audit Support	Audit Logs and Compliance Support Documents	Audit trail reports, access logs, configuration change logs, transaction logs, bid activity logs, invoice logs, payment logs, reconciliation logs, compliance records, and support for internal/external audits for both business modules and common platform services. The deliverables shall support auditability of REIA billing, exchange trading, bilateral trading, settlement, ledger, compliance reporting, and management dashboard activities.	Availability of records and support during audit/review
O&M Transition	O&M Handover and Support Plan	Handover of production system, support process, SLA tracking mechanism, escalation matrix, issue reporting process, configuration records, integration records, dashboard/report catalogue, known issues, release baseline, user access matrix, and transition to O&M phase for both modules and common services. O&M handover shall include support procedures for exchange integrations, bid workflows, billing/settlement, payment reconciliation, client ledger, digital signature/e-invoicing, price forecasting, dashboards, reports, SLA tracking, and escalation process.	O&M readiness confirmation by SJVN

7. Implementation Assumptions and Dependencies

The implementation of the proposed platform shall be subject to the following assumptions and dependencies:

- SJVN shall provide timely access to required business users, process owners, finance users, trading users, management users, and technical stakeholders for requirement finalization, reviews, UAT, and sign-offs.
- SJVN shall provide required master data, contract data, PPA data, NOC data, client data, historical bid data, exchange data, schedule data, billing data, payment data, and reconciliation data in agreed formats.
- Integration with external systems such as power exchanges, NOAR, NLDC, RLDC, SLDC, ERP/accounting systems, banking systems, digital signature/e-invoicing systems, email/SMS gateways, and other platforms shall be subject to availability of APIs/SFTP/interfaces, access credentials, technical documentation, permissions, approvals, and support from respective system owners.

- Where direct system integration is not available or not feasible, the system shall support manual upload/download through agreed templates or file-based mechanisms.
- Price forecasting scope, methodology, data sources, model complexity, refresh frequency, accuracy expectations, and output formats shall be finalized during the requirement gathering and design phase.
- Digital signature, e-invoicing, bank integration, and virtual account-based collection shall be implemented subject to availability of required third-party services, APIs, credentials, and approvals.
- Any major change in statutory, regulatory, exchange, market, tax, accounting, or business requirements after requirement sign-off shall be managed through a formal Change Request process.
- Timelines for implementation, integration testing, UAT, and go-live shall be dependent on timely availability of required data, interfaces, approvals, infrastructure, third-party coordination, and SJVN sign-offs.

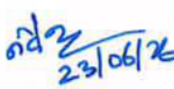
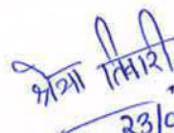
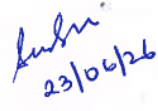
8. Acceptance and Sign-off

Each milestone and deliverable shall be considered complete only after submission by the bidder, review by SJVN, resolution of observations, and formal acceptance/sign-off by SJVN.

The following shall be treated as key acceptance points:

- Requirement sign-off.
- Functional and technical design sign-off.
- Completion of development and configuration.
- Completion of system integration testing.
- Completion of user acceptance testing.
- Go-live readiness approval.
- Successful production deployment.
- Completion of stabilization period.
- O&M transition sign-off.

SJVN may provide observations or comments on submitted deliverables, and the bidder shall address such observations within mutually agreed timelines. UAT sign-off and go-live approval shall be mandatory before production rollout of the respective module or functionality.

 23/06/26	 23/06/26	 23/06/26		 23/06/2026
N.N. Rahul	Shreya Tewari	Surbhi Kabra	Vikas Garg	Divyankur Choudhary
Sr. Manager (C&P)	Sr. Manager (IT)	Sr. Manager (PT&BDE)	Sr. Manager (F&A)	Sr. Manager (REIA)

